

## CSA5112 - Cryptography and Network Security

### 8. DES (Data Encryption Standard)

**Aim:** To implement the Data Encryption Standard (DES) algorithm for encrypting and decrypting a plaintext message using a secret key.

**Algorithm:**

1. Start.
2. Import the DES library.
3. Read the plaintext.
4. Create an 8-byte DES key.
5. Encrypt the plaintext.
6. Display the ciphertext.
7. Decrypt the ciphertext.
8. Display the original plaintext.
9. Stop.

**Code:**

```
from Crypto.Cipher import DES

from Crypto.Util.Padding import pad, unpad

import base64

key = b'12345678'    # 8-byte key

cipher = DES.new(key, DES.MODE_ECB)

plaintext = input("Enter Plain Text: ")

encrypted = cipher.encrypt(pad(plaintext.encode(), DES.block_size))

ciphertext = base64.b64encode(encrypted).decode()

print("Encrypted Text:", ciphertext)

decrypted = unpad(cipher.decrypt(encrypted), DES.block_size)

print("Decrypted Text:", decrypted.decode())
```

### Sample Input:

```
Enter Plain Text: HELLO
```

### Sample Output:

```
Enter Plain Text: HELLO  
  
Encrypted Text: 8iP8+xy8a4M=  
Decrypted Text: HELLO
```

### Result:

The DES algorithm was successfully implemented. The plaintext was encrypted and decrypted successfully using DES.